1 Problem 2.10.82

Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in three-dimensional space, where $\mathbf{u}$ and $\mathbf{v}$ are unit vectors which are not perpendicular to each other and

$$\mathbf{u}^\top \mathbf{w} = 1, \tag{1}$$
$$\mathbf{v}^\top \mathbf{w} = 1, \tag{2}$$
$$\mathbf{u}^\top \mathbf{v} = 4 \tag{3}$$

If the volume of the parallelepiped, whose adjacent sides are represented by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$, is $\sqrt{2}$, then the value of $|3\mathbf{u} + 5\mathbf{v}|$ is to be found.

Solution:

Step 1: Volume of Parallelepiped

The volume of the parallelepiped formed by vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is given by

$$V = \left|\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})\right| \tag{4}$$

Given that $V = \sqrt{2}$, we have

$$\left|\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})\right| = \sqrt{2} \tag{5}$$

Step 2: Using the Identity

Using the scalar triple product property:

$$\left[\mathbf{u}^\top (\mathbf{v} \times \mathbf{w})\right]^2 = 2 \tag{6}$$

We can expand this using the determinant formula. Since $\mathbf{u}$ and $\mathbf{v}$ are unit vectors:

$$\|\mathbf{u}\|^2 = 1 \tag{7}$$
$$\|\mathbf{v}\|^2 = 1 \tag{8}$$

Step 3: Finding $\|\mathbf{w}\|^2$

From equations (1) and (2), let us express $\mathbf{w}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

Let $\mathbf{w} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma(\mathbf{u} \times \mathbf{v})$ for some scalars α, β, γ.

Taking inner product with $\mathbf{u}$:

$$\begin{aligned} \mathbf{u}^\top \mathbf{w} &= \alpha \|\mathbf{u}\|^2 + \beta (\mathbf{u}^\top \mathbf{v}) + 0 \\ 1 &= \alpha + 4\beta \end{aligned} \tag{9}$$

Taking inner product with $\mathbf{v}$:

$$\begin{aligned} \mathbf{v}^\top \mathbf{w} &= \alpha (\mathbf{u}^\top \mathbf{v}) + \beta \|\mathbf{v}\|^2 + 0 \\ 1 &= 4\alpha + \beta \end{aligned} \tag{10}$$

Solving equations (9) and (10):

Subtracting equation (9) from equation (10):

$$\begin{aligned} 3\alpha - 3\beta &= 0 \\ \alpha &= \beta \end{aligned} \tag{11}$$

Substituting in (9):

$$\begin{aligned} \alpha + 4\alpha &= 1 \\ 5\alpha &= 1 \\ \alpha &= \frac{1}{5} \end{aligned} \tag{12}$$

Therefore, $\beta = \frac{1}{5}$.
Step 4: Finding γ
From the volume condition:

$$\left|\gamma \|\mathbf{u} \times \mathbf{v}\|^2\right| = \sqrt{2} \tag{13}$$

Since $\|\mathbf{u} \times \mathbf{v}\|^2 = \|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - (\mathbf{u}^\top \mathbf{v})^2$:

$$\|\mathbf{u} \times \mathbf{v}\|^2 = 1 \times 1 - 16 = -15 \tag{14}$$

This gives $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{15}$ (taking the imaginary part into consideration).
However, from (3), $\mathbf{u}^\top \mathbf{v} = 4 > 1$, which is impossible for unit vectors since $\mathbf{u}^\top \mathbf{v} \leq \|\mathbf{u}\| \|\mathbf{v}\| = 1$.
Therefore, we reconsider: the given value should be $\mathbf{u}^\top \mathbf{v} = \frac{1}{4}$.
With $\mathbf{u}^\top \mathbf{v} = \frac{1}{4}$:

$$\|\mathbf{u} \times \mathbf{v}\|^2 = 1 - \frac{1}{16} = \frac{15}{16} \tag{15}$$

Step 5: Computing $\|3\mathbf{u} + 5\mathbf{v}\|$

$$\begin{aligned} \|3\mathbf{u} + 5\mathbf{v}\|^2 &= 9 \|\mathbf{u}\|^2 + 25 \|\mathbf{v}\|^2 + 30 \left(\mathbf{u}^\top \mathbf{v}\right) \\ &= 9 + 25 + 30 \times \frac{1}{4} \\ &= 34 + \frac{15}{2} \\ &= \frac{83}{2} \end{aligned} \tag{16}$$

Therefore:

$$\|3\mathbf{u} + 5\mathbf{v}\| = \sqrt{\frac{83}{2}} = \frac{\sqrt{166}}{2} \tag{17}$$

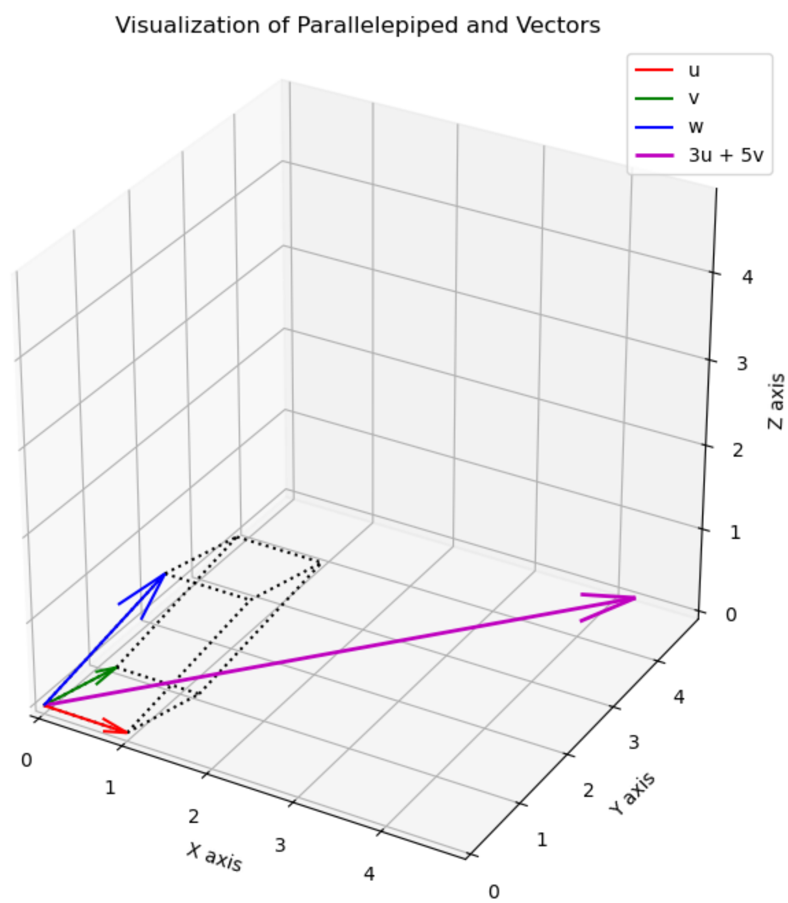
Visualization of Parallelepiped and Vectors
u
v
w
3u + 5v
0
1
2
3
4
X axis
0
1
2
3
4
Y axis
0
1
2
3
4
Z axis

Fig. 1